Oppg 1

a)

state 1 $(\vec{X}_1) = \begin{bmatrix} 100 \cdot 10^3 \\ 27+273,15 \\ 0,4 \end{bmatrix} \vec{b}$, state 2 $(\vec{X}_2) = \begin{bmatrix} 300 \cdot 10^3 \\ x_2 \\ 0,4 \end{bmatrix} \vec{b}$, $\Delta \vec{X}_{1-2} = \begin{bmatrix} 200 \cdot 10^3 \\ x_2 - 300,15 \\ 0 \end{bmatrix} \vec{b}$

enhetene $(\vec{b}) = \begin{bmatrix} [Pa] \\ [k] \\ [m^3] \end{bmatrix}$

$PV = TRm$

energiendring inn $(\Delta E) = \Delta Q + \Delta W$, $(\Delta W = 0$ isokor$)$

ΔQ: termisk energi; ΔW: bevegelsesenergi

Der temperaturen er mer enn 9°C, så antar jeg ideal gass.

$\frac{V}{mR} = \frac{T}{P} \rightarrow \frac{T_1}{P_1} = \frac{T_2}{P_2} \rightarrow T_2 = \frac{T_1}{P_1} P_2 = \frac{(27+273,15)\ 300 \cdot 10^3}{100 \cdot 10^3} = 900,45\ [k]$

($\frac{V}{mR}$: constant)

$\vec{X}_2 = \begin{bmatrix} 300 \cdot 10^3 \\ 900,45 \\ 0,4 \end{bmatrix} \vec{b}$

b)

state 3 $(\vec{X}_3) = \begin{bmatrix} 300 \cdot 10^3 \\ 1100 \\ x_3 \end{bmatrix}$, $\Delta \vec{X}_{2-3} = \begin{bmatrix} 0 \\ 199,55 \\ x_3 \end{bmatrix}$

$PV = mRT$

$\downarrow$

$m = \frac{PV}{RT} \rightarrow m = \frac{100 \cdot 10^3 \cdot 0,4}{287 \cdot 300,15} \approx 0,464\ [kg]$

isobar

$\Delta E = \Delta Q + \Delta W$, $\Delta W \neq 0$

sjennomsnittstemp $(\bar{T}) = \frac{T_1 + T_2}{2} = \frac{1400}{2} = 700\ [k]$

$c_v = 0,788$, $c_p = 1,075$

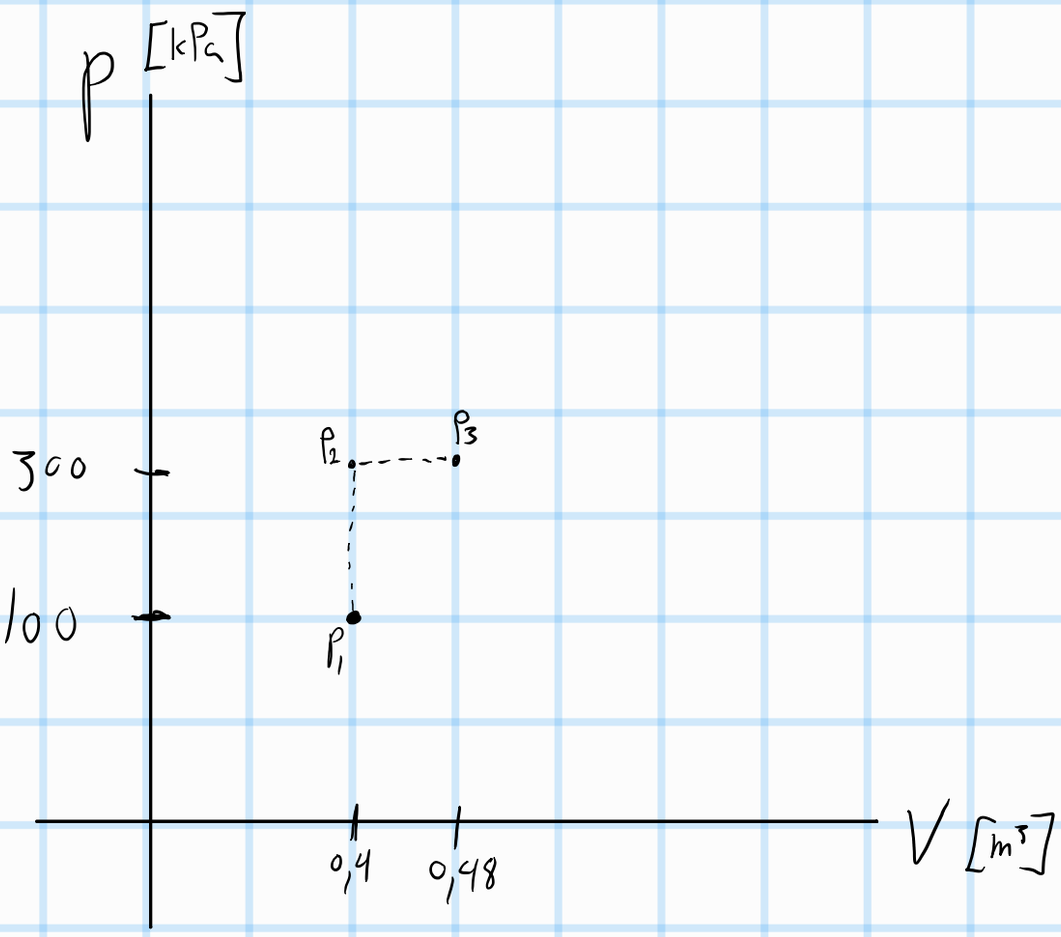
$Q = Q_{1-2} + Q_{2-3} = m\left(c_v (T_2 - T_1) + c_p (T_3 - T_2)\right)$, $m = \frac{100 \cdot 10^3 \cdot 0,4}{287 \cdot 300,15}$

$$\frac{100 \cdot 10^3 \cdot 0{,}4}{287 \cdot 300{,}15}\left(0{,}788\left(900{,}45 - 300{,}15\right) + 1{,}075\left(1100 - 900{,}45\right)\right)$$

$$319{,}2587\,(kJ)$$

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c)



oppg 2

trykk inn (p_1) = $100 \cdot 10^3$ [Pa]

trykk ut (p_2) = $700 \cdot 10^3$ [Pa]

temp v. inn (T_1) = 300 [k]

temp v. ut (T_2) = 600 [k]

massestrømmen ($\dot{m}$) = 0,25 [kg/s]

varmetapet ($\frac{dQ}{dm}$) = $15 \cdot 10^3$ [J/kg]

a)

volumstrømmen ($\dot{V}$) = $\dot{m} \cdot v_1$

spesifikk volum (v_1) = $\frac{R \cdot T_1}{p_1} = \frac{287 \cdot 300}{100 \cdot 10^3} = 8,61$

$\dot{V} = 0,25 \cdot 8,61 = 2,1525$ [m^3/s]

b)

i) konstant spesifik varmekapasitet

entalpiendring (Δh) = $h_2 - h_1 = c_p (T_2 - T_1)$

$c_p = 1,039$ $\qquad$ $600 - 300 = 300$

$1,039 \cdot 300 = 311,7 \cdot 10^3$ [J/kg]

arbeid inn ($\dot{W}_{inn}$) = $\left(\Delta h + \frac{dQ}{dm}\right) \cdot \dot{m}$ = $(311,7 + 15)\ 10^3 \cdot 0,25 = 86,675 \cdot 10^3$ [J/s] ([W])

ii)

$$\Delta h = h_2 - h_1$$

nitrogen molar mass $(M_{N_2}) = 28,013 \left[kg/mol \right]$

entring av entalpien per molarmasse $\left(\# \right) = \frac{\Delta h}{M_{N_2}}$

$$\Delta \# = \#_2 - \#_1 = 17,563 - 8,723 = 8,84 \left[kJ/mol \right]$$

$$\Delta h = \# \; M_{N_2} = 8,84 \cdot 28,013 = 247,63 \left[kJ/kg \right]$$

$$\dot{W}_{inn} = \left(\Delta h + \frac{dQ}{dm} \right) \dot{m} = \left(247,63 + 15 \right) 0,25$$

$$65,65873 \left[kJ/s \right]$$

$$(kW)$$

oppgave 3

$p_1 = 5 \cdot 10^6$ [Pa], $T_1 = 500 + 273{,}15$ [K], $V_1 = 20$ [m/s]

$p_2 = 30\ 10^3$ [Pa], $x = 0{,}95$, $V_2 = 80$ [m/s], $\dot{m} = 5$ [kg/s]

a)

endringen i dampens spesifike kinetiske energi $(ke) = \frac{v^2}{2}$

v = hastighet til vanndampen

$$\Delta ke = \frac{V_2^2 - V_1^2}{2} = \frac{80^2 - 20^2}{2} = 3000\ [m^2/s^2] = \underline{\underline{3000\ [J/kg]}}$$

$$[J] = [N] \cdot [m] = [kg] \cdot [m^2/s^2]$$

$$[J/kg] = [m^2/s^2]$$

b)

turbinens effekt ut $(\dot{W}_{ut}) = \dot{m}(h_1 - h_2 + ke)$

entalpi til superhetet vanndamp ved P_1 og T_1 $(h_1) = 3434{,}7\ [kJ/kg]$

entalpien til vanpamp ved P_2 og x $(h_2) = 289{,}27 + 0{,}95 \cdot 2335{,}3$

$= 2307{,}805\ [kJ/kg]$

$$5(3434{,}7 - 2307{,}805 + 3000)$$

$$\underline{\underline{20634{,}475\ [kW]}}$$

c)

$$\dot{V} = \dot{m} \cdot v_s$$

$$v_s = 0{,}039448 \left[m^3/kg \right]$$

$$\dot{V} = 0{,}19724 \left[m^3/s \right]$$

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d)

$$\text{nastiz hetven}\left(v_1 \right) = \frac{\dot{V}}{A_1}$$

$$A_1 = \frac{\dot{V}}{v_1} = \frac{0{,}19724}{20} = 9{,}862 \cdot 10^{-3}$$

$$9{,}862 \cdot 10^{-3} \left[m^2 \right]$$

oppgave 4

$$V = 25 \cdot \left(10^{-1}\right)^3 \, [m^3] \, , \quad \rho = 850 \, [kg/m^3]$$

$$C_{olje} = 2{,}2 \, [kJ/(kg \cdot °C)] \, , \quad T_1 = 12 \, [°C] \, , \quad T_2 = 55 \, [°C]$$

$$V_{luft} = 45 \, [m^3] \, , \quad P_1 = 100 \cdot 10^3 \, [Pa] \, , \quad T_{1luft} = 12 \, [°C]$$

$$T_{2Luft} = 22 \, [°C] \, , \quad \dot{W}_{el} = 1{,}5 \, [kW] \, , \quad \dot{Q}_{ut} = 0{,}25 \, [kW]$$

$$\left(\dot{W}_{El} - \dot{Q}_{ut}\right) t = \Delta U_{luft} + \Delta U_{olje}$$

$$m_{luft} = \frac{P_1 V_{luft}}{RT} = \frac{100 \cdot 10^3 \cdot 45}{287 \cdot 285{,}5} \approx 54{,}99 \, [kg]$$

$$\Delta U_{luft} = m_{luft} \cdot C_{v\,luft} \cdot \Delta T_{luft}$$

$$54{,}99 \cdot 0{,}718 \cdot 10 \approx 394{,}8 \, [kJ]$$

$$m_{olje} = \rho \cdot V_{olje} = 850 \cdot 0{,}025 = 21{,}25 \, [kg]$$

$$\Delta U_{olje} = m_{olje} \cdot C_{olje} \cdot \Delta T_{olje}$$

$$21{,}25 \cdot 2{,}2 \cdot 43 = 2010{,}3 \, [kJ]$$

$$\frac{\Delta U_{wft} + \Delta U_{olje}}{\dot{W}_{el} - \dot{Q}_{ut}} = t$$

$$354,8 + 2010,3 = 2405,1\ [kJ]$$

$$1,5 - 0,25 = 1,25$$

$$\frac{2405,1}{1,25} = 1924,1\ [s]$$

$$\approx 32,1\ \text{minutt}$$